

SELENE: Selective Empathy LoRA with Factor Gates for English–Korean Affective Transfer

Jaehong Kim, Seohyeon Yoo, and Jaehyuk Cho

Abstract—Large language models that perform well on English empathetic dialogue often fail to transfer that competence to Korean, where empathy is shaped by relational stance, honorifics, and high-context support strategies. We propose SELENE (Selective Empathy LoRA with EN–KO factor gatEs): a parameter-efficient EN→KO affective transfer system on a frozen Qwen/Qwen3.5–9B backbone. SELENE (i) pretrains an English empathy LoRA with an affect head on EmpatheticDialogues, (ii) assigns factor-wise share/relearn/suppress gates from EN–KO divergence and frozen probes, and (iii) adapts a Korean LoRA with multitask affect, strategy, and relation supervision on AI Hub dialogues. Unlike language×task adapter merging alone, SELENE gates theoretically motivated empathy factors (Affect / Cognition / Strategy / Relation). We evaluate two directions: Direction I Korean A/S/R accuracy versus non-selective baselines, and Direction II return-to-English performance after KO adaptation (gain versus forgetting), with KoED held out for cultural evaluation. Illustrative comparisons suggest that Affect/Cognition are largely shareable while Strategy/Relation require relearning—motivating selective gates over blind EN adapter copy.

Index Terms—Affective computing, empathetic dialogue, cross-lingual transfer, LoRA, cultural adaptation, Korean NLP, support strategies.

I. INTRODUCTION

EMPATHETIC response generation is a core affective computing problem that couples emotion understanding with socially appropriate generation [1], [2], [13]. Open-domain empathy benchmarks such as EmpatheticDialogues (ED) [1] and emotional-support corpora such as ESC [2] have driven rapid progress in English, and large language models (LLMs) further improve fluency on affective tasks [4], [12]. Yet multilingual competence does not imply multicultural empathy [3], [17], [18]: models that score well in English frequently violate Korean relational norms, over-empathize, or select support strategies that are pragmatically odd in honorific, high-context settings [3], [11].

A common remedy is full fine-tuning on Korean dialogues. This is compute-heavy and risks catastrophic forgetting of English affective skills [9], [22]. Parameter-efficient fine-tuning (PEFT), especially LoRA [5] and modular adapters [6],

[21], reduces cost and keeps the backbone frozen. However, typical EN→KO transfer still treats empathy as a *monolithic* skill: either copy an English adapter blindly, train Korean from scratch, or merge language×task adapters without asking which *empathy factors* should be shared versus relearned [7], [8].

Psychological and dialogue research argues that empathy is multidimensional [10]: affective sharing, perspective-taking, helping strategies, and culturally regulated display rules are separable. ESC further shows that support *strategy*—not emotion mirroring alone—drives perceived helpfulness [2], [13]. KoED demonstrates a measurable cultural empathy gap between ED and culturally reconstructed Korean dialogues [3]. Together, these observations motivate *factor-selective* transfer rather than uniform adapter reuse.

This paper asks: can we transfer English empathy to Korean selectively at the level of theoretically motivated affective factors, while keeping the backbone frozen and preserving return-to-English performance?

Our contributions are:

- 1) A theory-aligned A/C/S/R factor schema for empathetic dialogue (Affect, Cognition, Strategy, Relation), anchored in empathy theory, ESC strategies, and Korean relational norms [2], [3], [10], [11].
- 2) **SELENE**, implementing Factor-LoRA SELECT: English LoRA pretraining, divergence/probe gating (share/relearn/suppress), and gated Korean multitask LoRA adaptation on frozen Qwen/Qwen3.5–9B.
- 3) A dual evaluation protocol—Direction I (EN→KO affective accuracy) and Direction II (return-to-EN after KO)—plus stage-wise ablations that isolate EN pretraining, gates, and multitask heads.
- 4) A unified EN–KO data pipeline over EmpatheticDialogues, KoED, KorEmpatheticDialogues, and AI Hub under one JSONL schema, with factor-wise EN–KO comparison figures that justify selective gates.

II. RELATED WORK

A. Empathetic and Emotional Support Dialogue

EmpatheticDialogues remains the standard open-domain empathy benchmark for emotion-grounded response generation [1]. Subsequent systems emphasize mixture-of-experts emotion routing [14], emotion mimicking [15], and commonsense-aware empathy [16]. Emotional Support

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Conversation (ESC) shows that staged helping strategies (exploration→comforting→action) improve support effectiveness beyond affect mirroring [2]. Computational empathy metrics for support text further highlight explorations, interpretations, and emotional reactions as complementary axes [13]. EmoBench evaluates multi-faceted emotional intelligence in LLMs [4], and broad TAFFC evaluations report uneven LLM behavior across affective tasks [12]. Our Strategy (S) axis follows ESC and is operationalized with AI Hub labels (advice / encourage / comfort / agree), while Affect (A) uses emotion labels from ED and AI Hub.

B. Cultural Empathy and Multilingual Evaluation

KoED is our problem mother paper [3]: multilingual LLMs underperform on culturally reconstructed Korean dialogues relative to ED (emotion Acc 36.84 → 32.69 on shared emotions), and cultural appropriateness favors Korean-aligned models. This aligns with broader evidence that LLMs exhibit Western-centric cultural defaults even when prompted in other languages [17], and that multilingual benchmarks must separate linguistic fluency from cultural competence [18], [19]. Cross-cultural psychology motivates Relation (R) through display rules and emotion regulation under social context [11]. SELENE does not replace culturally aligned pretraining; instead it provides a PEFT mechanism to *selectively* adapt English empathy factors toward Korean A/S/R supervision while freezing a shared backbone.

C. Parameter-Efficient Cross-Lingual Transfer

Adapter modules [21] and LoRA [5] enable low-rank task adaptation without updating all backbone weights. MAD-X separates language and task adapters for modular cross-lingual transfer [6]. AdaMergeX merges task/language LoRAs with adaptive arithmetic and reports large gains over MAD-X-style baselines [7]. FLARE fuses EN/target representations inside LoRA bottlenecks and improves target-language accuracy over plain LoRA [8]. Strong multilingual encoders such as XLM-R underpin much of this transfer stack [20]. On the forgetting side, classical continual-learning results warn that sequential adaptation can erase prior skills [22], and LF-MLF shows that reducing source-side forgetting improves retained multilingual transfer [9]—motivating our Direction II return-to-EN protocol. These methods operate at language×task granularity; SELENE instead gates empathy factors A/S/R with share/relearn/suppress decisions.

III. PROBLEM FORMULATION

Let x be a dialogue context and y an empathetic listener response. We do *not* claim a unique latent decomposition of empathy. Instead we give operational definitions, testable propositions, and analytic consequences of the gate map that jointly justify factor-selective transfer.

A. Definitions

Definition 1 (Theory-aligned factors). Let $\mathcal{K} = \{A, C, S, R\}$. For each factor $k \in \mathcal{K}$, an observable label space $\mathcal{Y}_k$ is

fixed *a priori* by theory and corpus annotation (Table I): A = emotion, C = situation/appraisal proxy, S = support strategy, R = relation type [1]–[3], [10], [11]. When y_C is unavailable as a discrete label, C is treated as a proxy factor and excluded from gated decisions unless otherwise stated.

Definition 2 (Factor representation and head). Let $h_\theta(x) \in \mathbb{R}^d$ be the pooled hidden state of a frozen backbone (with optional EN LoRA). Define $z_k(x) := P_k h_\theta(x)$ with projection P_k (identity in the current prototype; factor subspaces / LoRA banks in the full system). A factor head f_k predicts $\hat{y}_k = f_k(z_k(x))$.

Definition 3 (Transfer score and gate map). On Korean labeled pairs $\{(x_i^{\text{KO}}, y_{k,i})\}$, the transfer score is

$$s_k = \text{Acc}(f_k^{\text{probe}}; h_\theta(x^{\text{KO}}), y_k^{\text{KO}}) \in [0, 1], \quad (1)$$

optionally mixed with an EN–KO alignment term. Fix thresholds $1 \geq \tau_{\text{share}} > \tau_{\text{suppress}} \geq 0$. The three intervals $[0, \tau_{\text{suppress}}]$, $(\tau_{\text{suppress}}, \tau_{\text{share}})$, $[\tau_{\text{share}}, 1]$ form a partition of $[0, 1]$, so there is a unique map

$$g_k = \Gamma(s_k) = \begin{cases} \text{share}, & s_k \geq \tau_{\text{share}}, \\ \text{suppress}, & s_k \leq \tau_{\text{suppress}}, \\ \text{relearn}, & \text{otherwise}, \end{cases} \quad (2)$$

sending each score to one gate. Equip gates with the total order $\text{suppress} \prec \text{relearn} \prec \text{share}$ (“more sharing” is larger).

Definition 4 (Gate-conditioned LoRA init). Let M_{EN} be the Stage 1 English LoRA and Rand a matched-rank random init. The init operator $\mathcal{I}$ maps a gate to a KO LoRA seed:

$$\mathcal{I}(g) = \begin{cases} M_{\text{EN}}, & g = \text{share}, \\ \text{Rand}, & g = \text{relearn}, \\ \rho M_{\text{EN}}, & g = \text{suppress}, \end{cases} \quad (3)$$

with scale $\rho \in [0, 1]$ (soft attenuation of EN directions). For factor set $\mathcal{K}$, Stage 3 seeds are $M_{\text{KO}}^{(0)} = \{\mathcal{I}(\Gamma(s_k))\}_{k \in \mathcal{K}}$.

We write $z = (z_A, z_C, z_S, z_R)$ as shorthand for Definition 2. English pretraining yields transferable components of z_A (and proxy z_C); Korean requires culture-specific realization especially for z_S and z_R [3]. The learning objective maximizes Korean A/S/R accuracy (Direction I) subject to limited degradation of English emotion recognition after KO adaptation (Direction II).

B. Propositions and Gate Lemmas

Proposition 1 (Separability). For each $k \in \{A, S, R\}$, a linear probe f_k on frozen $h_\theta(x^{\text{KO}})$ satisfies $\text{Acc}_k \geq \text{chance}_k + \delta$ for a pre-registered margin $\delta > 0$ (e.g., $\delta=0.1$), under a held-out KO split.

Proposition 2 (Partial non-redundancy). For $k \neq k'$, cross-factor probe accuracy $\text{Acc}(f_k \rightarrow y_{k'})$ is significantly lower than $\text{Acc}(f_k \rightarrow y_k)$, and/or Cramér’s $V(y_k, y_{k'})$ remains below a redundancy ceiling V_0 .

Proposition 3 (Transfer ordering). *Under an EN Stage 1 encoder, transfer scores obey $s_A \geq \tau_{\text{share}}$ and $s_S, s_R < \tau_{\text{share}}$ (equivalently $s_A, s_C \gg s_S, s_R$ when C is scored), supporting share for affect/(cognition) and relearn for strategy/relation.*

Proposition 4 (Gated intervention). *With matched backbone, rank, and budget, $\text{DirI}(\text{SELECT}) > \text{DirI}(\text{Blind})$ and $|\Delta_{\text{DirII}}(\text{SELECT})| \leq |\Delta_{\text{DirII}}(\text{Blind})|$, where Blind uses $\Gamma \equiv \text{share}$.*

Lemma 1 (Monotone gate and local stability). *Under Definition 3, Γ is (i) monotone nondecreasing w.r.t. $\prec$, and (ii) locally constant away from thresholds. Precisely: if $s \leq s'$, then $\Gamma(s) \preceq \Gamma(s')$. Moreover, if $\text{dist}(s, \{\tau_{\text{suppress}}, \tau_{\text{share}}\}) \geq \varepsilon > 0$, then every s' with $|s' - s| < \varepsilon$ satisfies $\Gamma(s') = \Gamma(s)$.*

Proof. Monotonicity follows because crossing a threshold can only move a score into a weakly higher gate region under $\prec$; within a region Γ is constant. For stability, the open ε -ball about s cannot reach either threshold, hence lies in the same partition cell as s , so Γ is unchanged. $\square$

Lemma 2 (Induced init is score-determined). *The Stage 3 seed map $s \mapsto \mathcal{I}(\Gamma(s))$ is uniquely determined by Definitions 3–4. If $\text{dist}(s_k, \{\tau_{\text{suppress}}, \tau_{\text{share}}\}) \geq \varepsilon$ and $|\hat{s}_k - s_k| < \varepsilon$, then $\mathcal{I}(\Gamma(\hat{s}_k)) = \mathcal{I}(\Gamma(s_k))$: bootstrap/probe noise that stays off the threshold boundaries does not flip the KO LoRA seed.*

Proof. Immediate from uniqueness of Γ on the partition and Lemma 1(ii), composed with the deterministic map $\mathcal{I}$ in (3). $\square$

C. Validation Protocol

Propositions 1–4 are *empirical claims*, not existence theorems. We verify them by: (i) KO held-out probes for (1); (ii) cross-factor Acc / Cramér’s V ; (iii) bootstrap tests of the ordering in Proposition 3; (iv) gated vs. blind / all-relearn interventions for Proposition 4; (v) seed/fold stability of $\{g_k\}$ and of $M_{\text{KO}}^{(0)}$ under Lemma 2. Lemmas 1–2 are analytic consequences of the definitions; they do *not* prove that A/C/S/R is the unique empathy factorization. Only after (iii)–(iv) hold do we apply $\{g_k\}$ via $\mathcal{I} \circ \Gamma$ in Stage 3.

IV. METHOD: SELENE (FACTOR-LoRA SELECT)

SELENE is the system name; Factor-LoRA SELECT is the three-stage algorithm. Unless noted, we freeze Qwen/Qwen3.5–9B—size-matched to KoED open peers (~ 7 –8B) [3]—and train only LoRA adapters plus lightweight A/S/R heads. Fig. 1 shows the pipeline; Fig. 2 shows the two evaluation directions; Fig. 3 summarizes the gate rule.

A. Theory-Based Factor Definition

Table I lists pre-defined factors and literature anchors. Divergence analysis is used to *validate* these factors, not to invent them post hoc from clusters. Davis separates empathic concern (A) from perspective-taking (C) [10]; Rashkin operationalizes emotion + situation in ED [1]; ESC supplies strategy supervision (S) [2]; Matsumoto and KoED motivate relational/cultural display constraints (R) [3], [11].

SELENE / Factor-LoRA SELECT pipeline

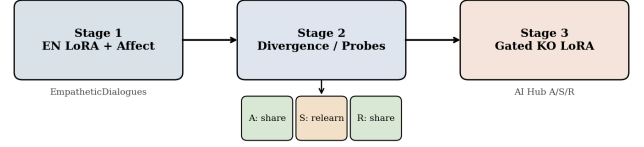


Fig. 1. SELENE overview: English empathy LoRA, factor-wise gates, then gated Korean multitask adaptation.

Evaluation protocol: Direction I / II

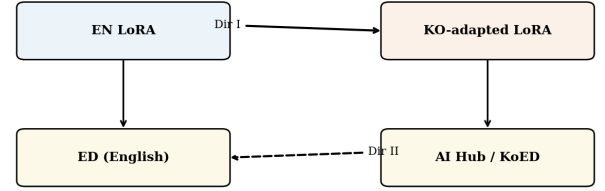


Fig. 2. Evaluation protocol. Direction I measures Korean A/S/R accuracy; Direction II returns to English ED after KO adaptation.

Factor gate rule (placeholder illustration)

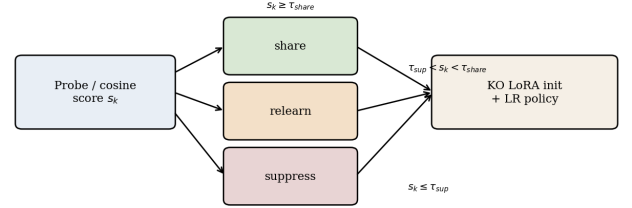


Fig. 3. Factor gate rule mapping probe/similarity scores to share / relearn / suppress (illustrative).

TABLE I
THEORY-ALIGNED EMPATHY FACTORS

Factor	Theory	Operational label
A Affect	Empathic concern [10]	Emotion (ED / AI Hub)
C Cognition	Perspective-taking [10]	Situation text (proxy)
S Strategy	ESC helping skills [2]	Advice / encourage / comfort / agree
R Relation	Display rules + KO norms [3], [11]	AI Hub relation types

B. Stage 1: English Empathy LoRA

On EmpatheticDialogues we train a causal LM with LoRA and an affect classification head on pooled prompt states:

$$\mathcal{L}_{\text{EN}} = \mathcal{L}_{\text{LM}} + \lambda_A \mathcal{L}_{\text{CE}}(A). \quad (4)$$

This yields a reusable English empathy adapter M_{EN} while leaving θ intact, consistent with PEFT practice [5].

C. Stage 2: Divergence / Probe Gates

Stage 2 instantiates Definitions 3–4: with the frozen Stage-1 encoder we (i) fit KO linear probes for A/S/R to obtain s_k in (1), (ii) optionally measure EN–KO / KoED alignment [3], and (iii) apply Γ in (2) (Lemmas 1–2). When English lacks explicit S/R supervision, we prefer **relearn** unless probe evidence

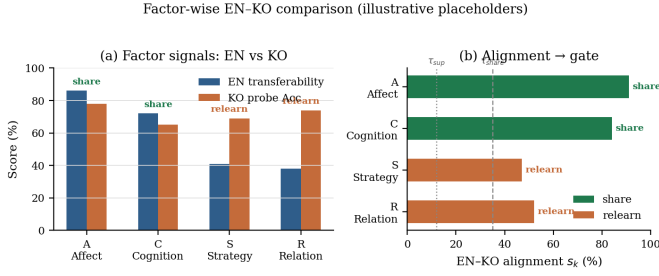


Fig. 4. Factor-wise EN-KO comparison (illustrative placeholders). (a) EN transferability vs. KO probe Acc for A/C/S/R. (b) EN-KO alignment s_k mapped to share/relearn gates.

TABLE II
FACTOR-WISE EN-KO PATTERN (ILLUSTRATIVE; DRIVES STAGE 2 GATES)

Factor	EN signal	KO probe	Align s_k	Gate
A Affect	high	high	0.91	share
C Cognition	mid-high	mid	0.84	share
S Strategy	low	high	0.47	relearn
R Relation	low	high	0.52	relearn

is strong. Fig. 4 and Table II summarize the intended EN-KO pattern corresponding to Proposition 3: Affect/Cognition tend to align (share), while Strategy/Relation diverge and require relearning.

D. Stage 3: Gated Korean Adaptation

On AI Hub dialogues we set $M_{KO}^{(0)} = \mathcal{I}(\Gamma(s_k))$ per factor (Definition 4): share copies M_{EN} (smaller LR to refine), relearn draws Rand, and suppress applies ρM_{EN} . We then train multitask objectives

$$\mathcal{L}_{KO} = \mathcal{L}_{LM} + \lambda_A \mathcal{L}_A + \lambda_S \mathcal{L}_S + \lambda_R \mathcal{L}_R, \quad (5)$$

so Korean adaptation is supervised not only for fluency but for A/S/R analysis heads, following the multi-axis empathy view in ESC and support-text evaluation [2], [13]. A composer weight α mixes $EN \oplus KO$ adapters for Direction II, preserving plug-in modularity [6], [7].

V. EXPERIMENTAL SETUP

A. Datasets and Backbone

Table III summarizes corpora. AI Hub provides full A/S/R labels; ED provides A/C; KoED is held out for cultural evaluation (not training), following [3]. KorEmpatheticDialogues serves as a translate-train control. All main comparisons freeze Qwen/Qwen3.5-9B and match LoRA rank across systems; training curves in Fig. 8 illustrate Stage 1/3 optimization dynamics.

B. Literature Anchor (Mother Paper)

Following KoED’s controlled ED $\leftrightarrow$ KoED comparison [3], Table IV reprints their reported emotion-recognition averages. This anchors Direction I: even strong multilingual models drop on culturally adapted Korean empathy, so “untouched multilingual” is an insufficient upper bound.

Algorithm 1 SELENE / Factor-LoRA SELECT

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Freeze backbone Qwen/Qwen3.5-9B
Stage 1: train EN LoRA + affect head on ED
Stage 2: encode EN/KO; assign gates  $g_k$ 
 $g_k \in \{\text{share, relearn, suppress}\}$ 
Stage 3: init KO LoRA from gates; train A/S/R
return EN LoRA, KO LoRA,  $\{g_k\}$ 

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TABLE III
DATASET STATISTICS

Corpus	Role	Split size
EmpatheticDialogues	EN Stage 1 / Dir II	17,844 / 2,763 / 2,542
AI Hub Empathetic Dial.	KO Stage 3 / Dir I	25,456 / 3,182
KoED	Cultural eval (held out)	1,360 (test)
KorEmpatheticDialogues	Translate-train baseline	19,531 / 2,769 / 2,547

TABLE IV
KoED EMOTION RECOGNITION ACC (%). REPRINTED AVERAGES FROM [3] (NOT OUR RUNS).

Model	ED (English)		KoED (Korean)	
	w/o j/h	w/ j/h	w/o j/h	w/ j/h
Llama-3.1-8B	34.53	30.74	32.19	28.53
Mistral-7B	39.38	36.40	33.52	31.10
Qwen2-7B	32.73	30.74	25.70	22.57
EXAONE-7.8B	31.41	29.12	30.31	26.18
Claude-3.5	46.17	42.94	41.72	38.31
Average	36.84	33.99	32.69	29.34

TABLE V
SYSTEMS COMPARED (DIRECTION I/II)

System	EN $\rightarrow$ KO policy	Dir I	Dir II
Untouched KO	KO LoRA from scratch (no EN init)	✓	✓
EN-only	Stage 1 EN LoRA on KO (no KO train)	✓	—
Blind share	Copy EN LoRA; no factor gates	✓	✓
Translate-train	KoED supervision	✓	—
MAD-X	Language + task adapters [6]	✓	—
AdaMergeX	Adaptive adapter merge [7]	✓	—
FLARE	EN/target LoRA fusion [8]	✓	—
SELENE (ours)	Factor gates share/relearn/suppress	✓	✓
+ compose α	EN $\oplus$ KO mix on return-to-EN	—	✓

C. Systems

Table V lists systems in a KoED-style checklist. Baselines span no-transfer (Untouched KO), zero-shot EN (EN-only), non-selective PEFT (Blind share), translation supervision, and strong modular XLT peers (MAD-X / AdaMergeX / FLARE) [6]–[8].

D. Direction I — Korean Affective Accuracy

Question: Does selective transfer improve Korean A/S/R over non-selective baselines? **Metrics:** Affect (A), Strategy (S), Relation (R) Acc on AI Hub valid; optional KoED CA (5-point) as in [3], [13]. Table VI and Fig. 5 follow KoED’s model $\times$ metric layout. *Numbers are illustrative placeholders pending full Qwen runs; best values in bold.* Under this placeholder regime, SELENE leads on Avg Acc and CA, with

TABLE VI

DIRECTION I: PERFORMANCE COMPARISON ON KOREAN EMPATHETIC ANALYSIS (ILLUSTRATIVE PLACEHOLDERS). PRIMARY METRICS ARE A/S/R ACC (%) ON AI HUB VALID; KoED CA IS 5-POINT.

System	AI Hub (Korean)				KoED
	A Acc $\uparrow$	S Acc $\uparrow$	R Acc $\uparrow$	Avg $\uparrow$	CA $\uparrow$
Untouched KO	52.1	48.5	44.0	48.2	2.94
EN-only	41.3	36.2	30.5	36.0	2.61
Blind share	55.4	54.0	46.8	52.1	3.12
Translate-train	53.0	50.1	45.2	49.4	3.05
MAD-X	56.2	55.5	48.1	53.3	3.18
AdaMergeX	57.8	56.9	49.0	54.6	3.27
FLARE	58.1	57.2	49.6	55.0	3.31
SELENE (ours)	61.4	63.8	54.7	60.0	3.55

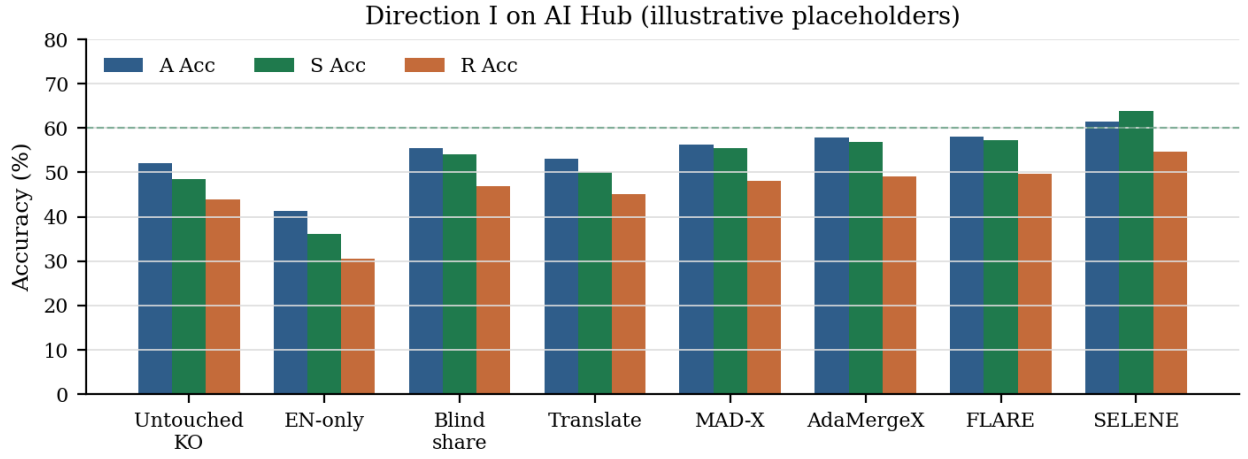


Fig. 5. Direction I A/S/R Acc on AI Hub (illustrative placeholders matching Table VI).

TABLE VII

DIRECTION II: RETURN-TO-EN ON EMPATHETIC DIALOGUES VALID (ILLUSTRATIVE PLACEHOLDERS). Δ IS RELATIVE TO EN BEFORE KO.

System (adapter on ED)	Acc $\uparrow$	Δ Acc	Δ LM
EN before KO	61.0	0.00	0.00
After Untouched KO	48.2	-12.8	+0.37
After Blind share	56.5	-4.5	+0.05
After SELENE	59.8	-1.2	+0.01
After SELENE + α	60.6	-0.4	+0.00

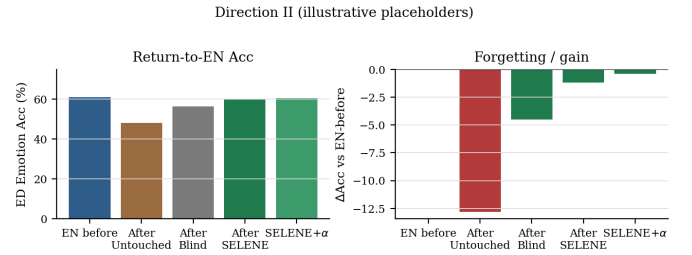


Fig. 6. Direction II return-to-EN Acc and Δ Acc (illustrative placeholders).

the largest relative lift on Strategy—consistent with ESC’s claim that strategy is the fragile transfer dimension [2].

E. Direction II — Return to English after KO

Question: After KO adaptation, is English empathy preserved? **Metrics:** ED emotion Acc; Δ Acc / Δ LM vs. EN-before-KO. Table VII and Fig. 6 use the same row-as-system style (illustrative placeholders). KO-scratch induces the largest English drop, while SELENE (and compose α) stays near the EN-before reference—echoing less-forgetting multilingual fine-tuning [9], [22].

F. Stage-Wise Ablation

Table VIII and Fig. 7 isolate each stage (illustrative placeholders). Removing Stage 1 (A1) or forcing all-relearn (A5) collapses Dir I and worsens Dir II; removing Stage 2 gates (A2) recovers Blind-share behavior; removing A/S/R heads (A3) hurts affective Acc even when LM loss is competitive—confirming multitask heads are not cosmetic [2]. Fig. 8 shows Stage 1/3 training curves.

G. Claims

- **Factor pattern:** A/C are EN-KO aligned (share); S/R are KO-specific (relearn)—Fig. 4, Table II.

TABLE VIII
STAGE-WISE ABLATION (ILLUSTRATIVE PLACEHOLDERS).

ID	Configuration	Dir I Avg $\uparrow$	Dir II Δ Acc	Isolates
A0	Full SELENE	60.0	−1.2	reference
A1	w/o Stage 1 (KO scratch)	48.2	−12.8	EN pretrain
A2	w/o Stage 2 (blind share)	52.1	−4.5	factor gates
A3	w/o Stage 3 heads (LM only)	41.0	−2.0	A/S/R heads
A4	Stage 1 only (no KO train)	36.0	0.0	KO adapt
A5	all-relearn	49.5	−11.5	share benefit
A6	all-suppress	50.8	−7.0	soft unlearn
A7	gate A only	55.2	−2.8	single factor
A8	compose α	59.1	−0.4	forgetting

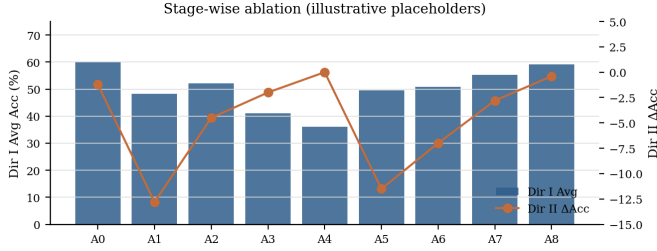


Fig. 7. Stage-wise ablation overview (illustrative placeholders).

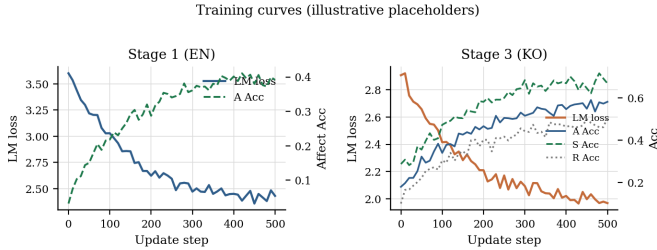


Fig. 8. Stage 1 (EN) and Stage 3 (KO) training curves (illustrative placeholders).

- **Dir I:** SELENE > Untouched KO and Blind share on A/S/R (esp. S/R), matching ESC’s emphasis on strategy [2].
- **Dir II:** SELENE does not incur larger English forgetting than Blind share; compose α further reduces forgetting [9].
- **Ablation:** removing Stage 2 (A2) or EN pretrain (A1) hurts Dir I; removing A/S/R heads (A3) hurts affective Acc beyond LM alone.

VI. DISCUSSION

SELENE addresses the cultural empathy gap documented by KoED [3] with a method complementary to language $\times$ task XLT [6]–[8]: factor-wise share/relearn/suppress on A/S/R while freezing the backbone. The EN–KO factor comparison (Fig. 4) is the mechanistic justification for selective gates rather than blind adapter copy: strategy and relation carry Korean interactional norms that English empathy adapters do not encode [3], [11], whereas affect and situation appraisal transfer more readily [1], [10]. This view is consistent with cultural-dominance findings in LLMs [17] and with multi-dimensional empathy theory [10], [13]. Direction II follows the less-forgetting intuition of LF-MLF [9] and classical

continual-learning warnings [22]. Tables VI–VIII and Figs. 5–8 currently use illustrative placeholders; full Qwen3.5-9B runs will replace them.

a) *Limitations.*: Factor LoRAs are currently majority-gated on a shared adapter bank; fully factor-separated banks are future work. Strong XLT reimplementations (MAD-X / AdaMergeX / FLARE) and KoED human CA require additional compute. AI Hub licensing constrains redistribution of raw dialogues. Placeholder tables/figures must not be read as submission-ready state-of-the-art rankings.

VII. CONCLUSION

We presented **SELENE**, a selective empathy LoRA system for English–Korean affective transfer on frozen Qwen3.5-9B. Across factor-wise EN–KO comparisons, Direction I/II evaluations, and stage ablations, one conclusion emerges: **empathy does not transfer as a single skill**—Affect and Cognition are largely shareable from English, whereas Strategy and Relation must be relearned for Korean [2], [3], [10]—so factor-gated adapters improve Korean A/S/R accuracy without the English forgetting induced by KO-scratch or blind overwrite [9]. Replacing the illustrative placeholders with full Qwen3.5-9B runs and KoED cultural appropriateness scores is the immediate next step.

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